

All the exercises should be performed in Jupyter Notebook.

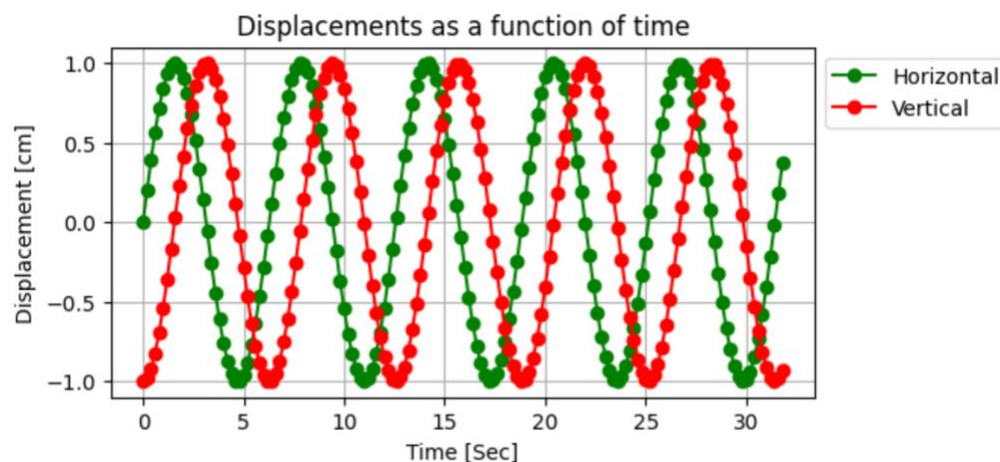
Exercise 1: Line plot and scatter plot with Matplotlib

Starting with the following code:

```
import matplotlib.pyplot as plt
import numpy as np

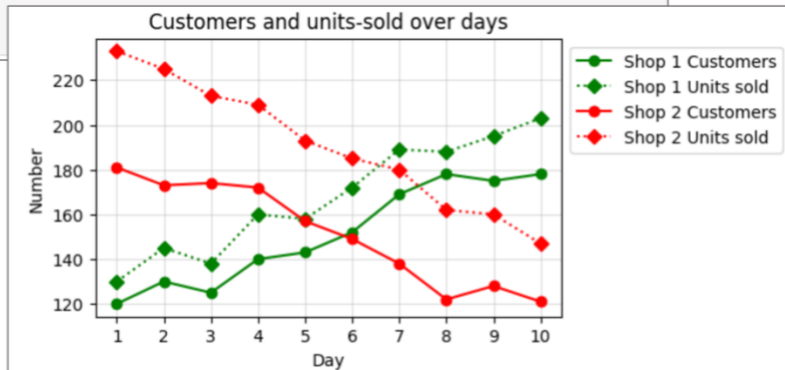
# Generating the data: time and displacements
time = [i / 5 for i in range(0,160)]          # time (in seconds)
displacement_h = [np.sin(a) for a in time]     # horizontal displacement [cm]
displacement_v = [-1*np.cos(a) for a in time]  # vertical displacement [cm]
```

(1a) Generate the following plot (including layout), using `plt.plot()`:

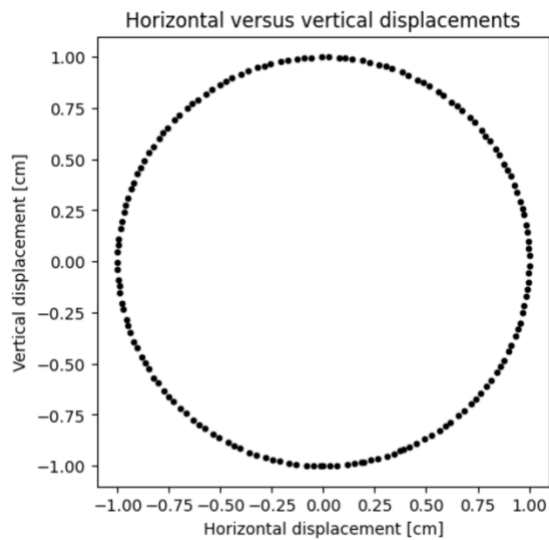


Hint, an example from today's lecture:

```
plt.figure(figsize=(5, 3))
plt.plot(day, shop_1_customers, marker = "o", c = "g", label="Shop 1 Customers")
plt.plot(day, shop_1_units_sold, marker = "D", c = "g", linestyle=':', label="Shop 1 Units sold")
plt.plot(day, shop_2_customers, marker = "o", c = "r", label="Shop 2 Customers")
plt.plot(day, shop_2_units_sold, marker = "D", c = "r", linestyle=':', label="Shop 2 Units sold")
plt.xticks(range(1, 11, 1))
plt.title("Customers and units-sold over days")
plt.xlabel("Day")
plt.ylabel("Number")
plt.grid(True, alpha = 0.4)
plt.legend(loc='upper left', bbox_to_anchor=(1, 1))
plt.show()
plt.close()
```



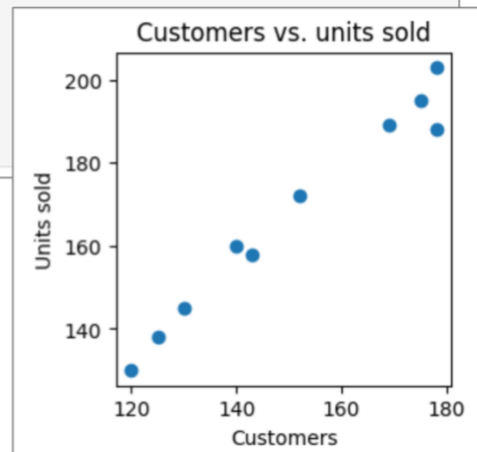
(1b) Generate the following plot (including layout), using `plt.scatter()`:



Hint, an example from today's lecture:

```
1 plt.figure(figsize=(3, 3))
2 plt.scatter(customers, units_sold) # Scatter plots of customers vs. units sold
3 plt.title("Customers vs. units sold")
4 plt.xlabel("Customers")
5 plt.ylabel("Units sold")
6 plt.show()
7 plt.close()
```

`plt.scatter()`



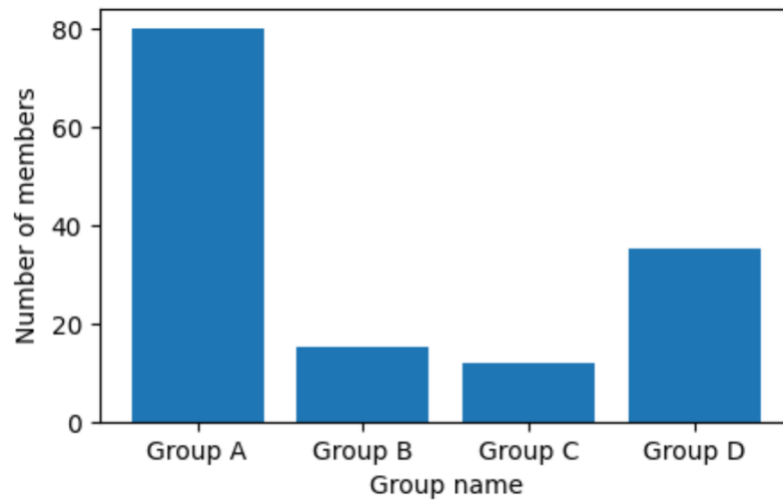
Exercise 2: Bar plot with Matplotlib

Starting with the following code:

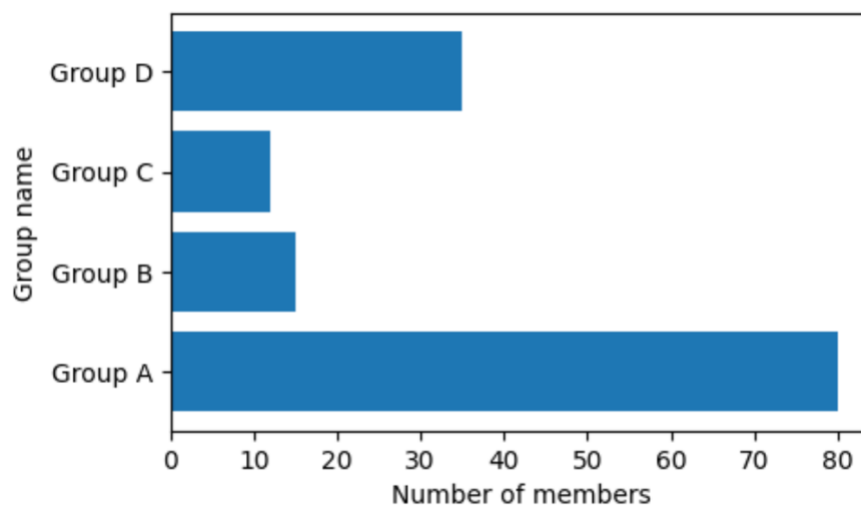
```
groups = ["Group A", "Group B", "Group C", "Group D"] # group name
members = [80, 15, 12, 35] # number of members
```

Generate the following bar plots:

(2a) Using `plt.bar()`:



(2b) Using `plt.barh()`:



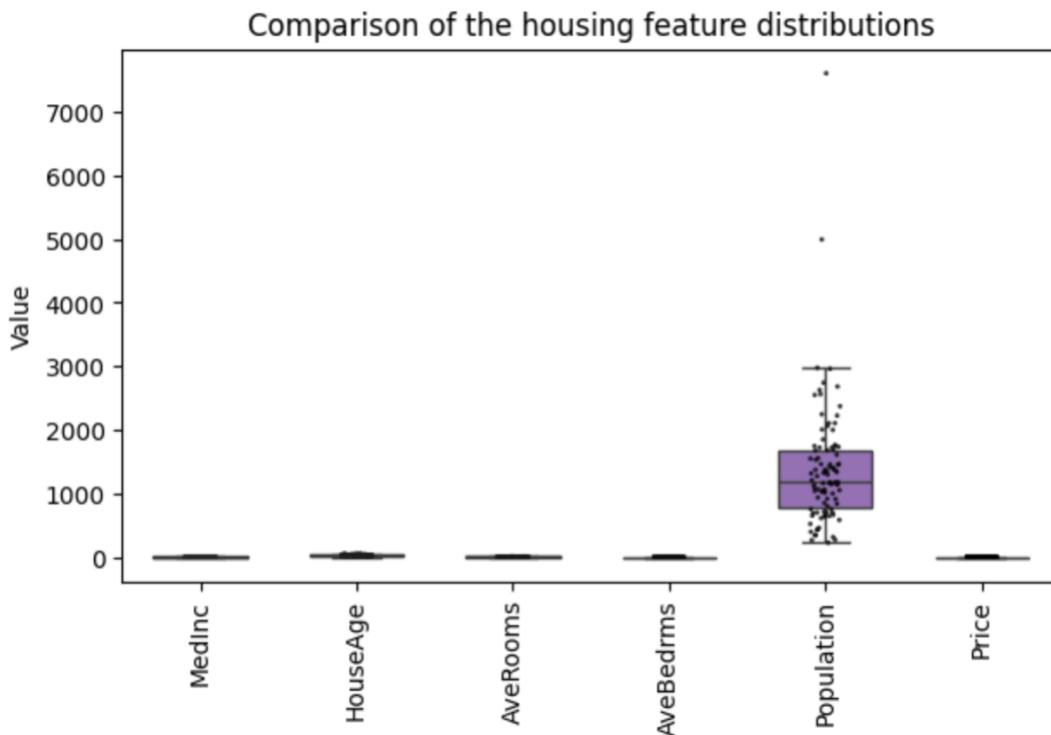
Exercise 3: Box plots with seaborn

Start with the following code to load a subset of the scikit-learn housing dataset (uploaded as a csv file “Data_sklearn_housing_subset.csv” on the Canvas board of the course) to a DataFrame:

```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
```

```
df_housing = pd.read_csv("Data_sklearn_housing_subset.csv", index_col = 0)
```

(3a) Using `sns.boxplot()` and `sns.stripplot()` generate the following plot:



Hint, an example from today's lecture:

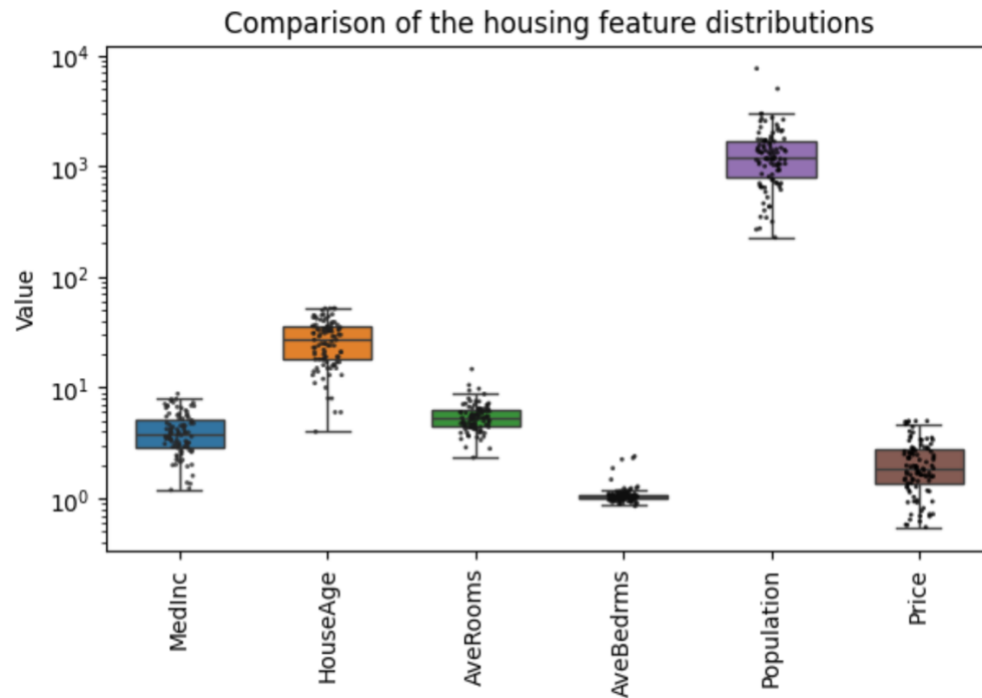
```
1 plt.figure(figsize=(7, 4))
2 sns.boxplot(data=wine_features, showfliers=False, width=0.6)
3 sns.stripplot(data=wine_features, palette='dark:black', alpha=0.8, legend=False, size = 2)
4 plt.ylabel("Value")
5 plt.xticks(rotation=90)
6 plt.title("Comparison of the wine feature distributions")
7 plt.show()
8 plt.close()
```

(3b) Check if all values in the DataFrame are > 0 using:

```
print((DataFrameName > 0).all().all())
```

You should get "True", meaning that one can apply directly a log scale for visualization.

So, generate the same plot as in 3a, but with a log scale for the y-axis, by adding the statement `plt.yscale('log')`:



(3c) Standardize the `df_housing` DataFrame, as was shown in today's lecture (for the wine dataset):

```
wine_features_standartized = (wine_features - wine_features.mean(axis=0)) / wine_features.std(axis=0)
```

And generate the same plot as in 3a, but with the standardized data:

